

Lesson 2.2: Maximin

Decision-making under uncertainty

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Fundraiser example

Example (Fundraiser)

A school has scheduled a fund-raiser a month into the future. It has a choice between holding a fête or a sports day. On a dry day, it expects to make \$150 on a fête and \$120 on a sports day. However, on a wet day it would make only \$75 on a fête and \$85 on a sports day.

	d	w
S	120	85
F	150	75

Maximin: choose based on best worst case scenario

- ▶ choose action(s) based on *best worst case*
- ▶ originally proposed by Abraham Wald
- ▶ for the extremely risk averse

	d	w	min
S	120	85	85
F	150	75	75

Example walk-through

	s_1	s_2	s_3	min
A_1	6	0	4	0
A_2	2	5	1	1
A_3	4	3	2	2

- ▶ *Maximin* would choose A_3
- ▶ similar rules in lecture notes:
 - ▶ *MaxiMax*
 - ▶ Hurwicz's rule